

Time resolution of the MX17 resistive Micromegas in micro-TPC mode

June 2026 cosmic bench — det3 (`sat_det3`) with det2 cross-check

MX17 cosmic-bench analysis (paper topic 10)

2026-07-11

Abstract

We measure the time resolution of the MX17 resistive-strip Micromegas detectors operated in micro-TPC mode on the June cosmic bench. Contrary to an earlier “not feasible” assessment, the bench *does* carry an absolute time reference: acquisition is triggered by a top+bottom scintillator-paddle coincidence, and the DREAM readout applies the 3-bit fine timestamp (`ftst`, 10 ns granularity) that re-references the free-running 60 ns sampling clock to the trigger. We verify the fine-timestamp correction directly in the data. Three complementary numbers follow, all on the frozen data set: (i) a single-strip time resolution of $\sigma_t = 38.9$ ns (1.33 mm of drift); (ii) a clean, telescope- and geometry-free *detector* resolution from the agreement of the two orthogonal readout layers, $\sigma_t = 33.1$ ns per plane (1.13 mm; 29.0 ns=1.00 mm after a residual time-walk correction), with an inter-plane bias consistent with zero (−1.3 ns); and (iii) an absolute event-time resolution (leading edge relative to the trigger) of $\sigma_{68} = 37.7$ ns, whose error budget is dominated by the detector term, not by the trigger. det2 reproduces every number within ~10%. The ≈ 1 mm-of-drift timing precision matches the transverse spatial resolution — the micro-TPC is not timing-limited relative to its geometry.

1 The question, and what “time resolution” means here

Hit mode answers *where* a muon crossed; micro-TPC mode adds *when* each drift electron arrived, which is what turns a single plane into a 3-D track segment. The relevant performance number is therefore the precision of the per-strip arrival time and of the event timestamp it defines. Two distinct questions hide under “time resolution”:

- **Absolute / time-of-flight:** how well does the detector timestamp the muon relative to an external clock? This needs a trigger reference.
- **Intrinsic / detector:** how reproducibly does the detector time an event using only its own channels? This is the number that limits micro-TPC tracking, independent of the trigger quality.

We measure both. The key enabler — and the correction to the earlier assessment — is that the absolute reference is present.

2 The timing chain: how an absolute time is formed

The per-strip time is reconstructed in `mm_strip_reconstruction` (`WaveformAnalyzer.cpp`) as a **constant-fraction** leading-edge crossing: the waveform is walked down from its peak to 30% of the baseline-subtracted amplitude and linearly interpolated between the two bracketing 60 ns samples, giving a sub-sample time (Fig. 1, right). Because the threshold tracks each pulse’s own amplitude,

this is largely free of amplitude time-walk by construction — a point we confirm quantitatively in Sec. 4.1.

The DREAM 60 ns sampling clock is *free-running* with respect to the trigger. When the scintillator coincidence fires, the readout window opens at whatever phase the clock happens to be in; that phase is recorded by the 3-bit fine timestamp `ftst` (0–7, 10 ns steps from the 100 MHz clock). The reconstruction **adds** `ftst` $\times$ 10 ns to every strip time (Fig. 1, left), re-referencing the window start to the trigger to 10 ns granularity. The result is that each strip’s `time` is an absolute leading-edge time relative to the (phase-corrected) trigger.

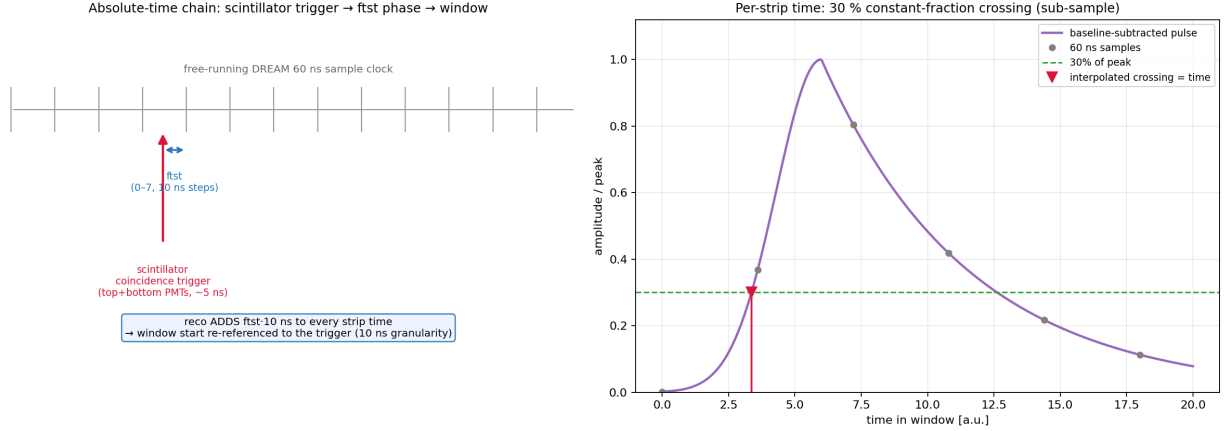


Figure 1: The absolute-time chain. *Left:* the scintillator-coincidence trigger arrives at a random phase of the free-running 60 ns DREAM clock; `ftst` records that phase (10 ns steps) and the reconstruction adds it back to every strip time. *Right:* the per-strip `time` is a 30%-of-peak constant-fraction crossing, linearly interpolated to sub-sample precision.

The fine-timestamp correction, verified in data. If the `ftst` correction were *not* applied, the reconstructed event time would slide by 10 ns per `ftst` unit. As applied, the leading-edge event time is flat in `ftst` (slope -0.2 ns/step); undoing the correction restores the expected -10.2 ns/step (Fig. 2, left). Correcting the phase also tightens the absolute event-time distribution from $\sigma_{68} = 41.3$ ns to 37.7 ns (Fig. 2, right). This is direct evidence that the sampling phase is measured and removed, i.e. that an absolute time exists.

3 Method: three observables

All observables use the detector’s own combined hits (spark/multiplicity veto ≤ 50 strips, hit amplitude > 100 ADC, pulse-start time in (0, 2000) ns). Per event and plane we take the largest spatial strip cluster (≥ 3 strips). Nothing here uses the M3 telescope — the measurement is self-contained. We adopt $v_{\text{drift}} = 34.30 \mu\text{m/ns}$ (the `sat_det3` geometric drift velocity at 1000 V) to translate a time to an equivalent drift distance, $\sigma_z = v_{\text{drift}} \sigma_t$.

A. Single-strip σ_t . For inclined tracks the core strips (amplitude $\geq 30\%$ of the cluster max, ≥ 4 strips spanning ≥ 2 mm) fall on a line time(position). The $(N-2)$ -normalised RMS of the residual is the per-strip timing precision.

B. Plane-to-plane detector σ_t . The X and Y strip layers sit under *one* drift gap and collect the *same* drifting electrons (Fig. 3), so the shortest-drift charge produces the same leading-

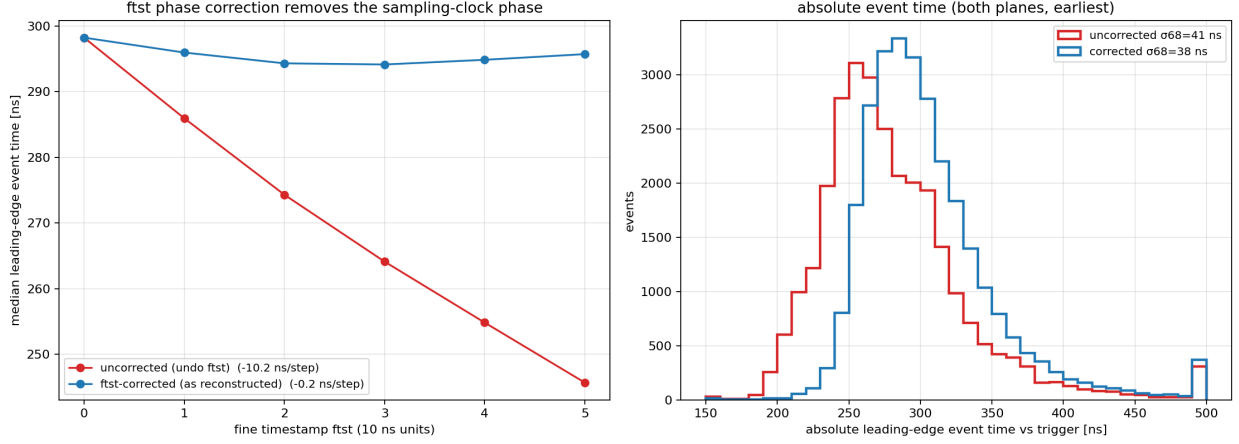


Figure 2: Fine-timestamp phase correction. *Left:* median leading-edge event time vs `ftst`; the applied correction (blue) is flat, the uncorrected case (red) walks by -10 ns/step. *Right:* the absolute event-time distribution is tighter with the correction applied.

edge time in both layers. The two layers are two *independent* measurements of one event timestamp: $\sigma(t_X - t_Y)/\sqrt{2}$ is the single-plane detector resolution, and the median of $t_X - t_Y$ is the residual inter-plane time-walk (a bias). Differencing cancels the trigger, the `ftst` phase, and the drift geometry, isolating the detector.

C. Absolute event-time σ_{68} . The earliest leading-edge time of the event (either plane), relative to the phase-corrected trigger. This is the time-of-flight-style number; it is an upper limit because it still contains the event-to-event variation of the minimum drift distance.

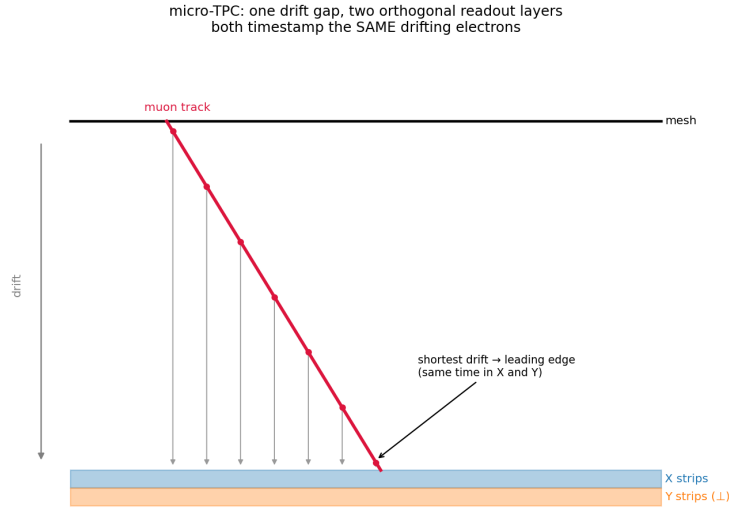


Figure 3: Micro-TPC geometry. A single drift gap and mesh above two orthogonal strip layers; every drift electron induces on an X and a Y strip at the same time, so the two planes independently time the same arrival sequence. The shortest-drift ionization sets the event leading edge.

4 Results (det3, sat_det3, 28 404 dual-plane events)

4.1 A. Single-strip time resolution

The per-strip residual has a median $\sigma_t = 38.9$ ns ($= 1.33$ mm of drift; $x/y = 39.2/38.6$ ns; Fig. 4, left). The per-strip *time-walk* — the residual as a function of the strip’s relative amplitude — is only $+3$ ns per unit relative amplitude (Fig. 4, right), i.e. negligible. This confirms that the 30% constant-fraction crossing already removes amplitude walk at the single-strip level, so no per-strip walk calibration is needed for the track fit.

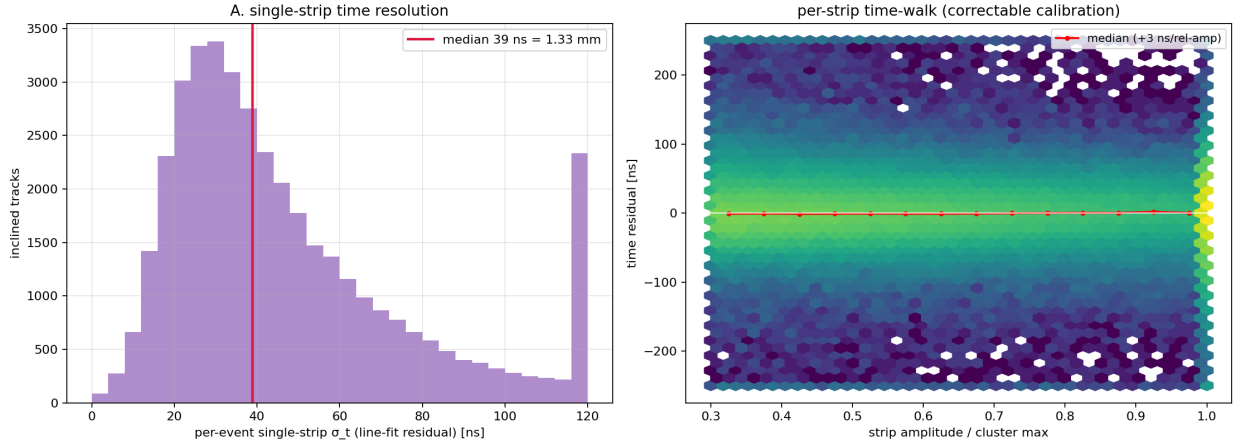


Figure 4: *Left:* single-strip time resolution from the micro-TPC line-fit residual. *Right:* per-strip time residual vs relative amplitude — the near-flat trend ($+3$ ns/rel-amp) shows the constant-fraction timing is walk-free at the strip level.

4.2 B. Plane-to-plane detector resolution (headline)

The two layers agree with an inter-plane bias of -1.3 ns — consistent with zero — directly confirming the “one gap, two orthogonal readouts of the same arrivals” picture. The spread gives a single-plane detector resolution of $\sigma_t = 33.1$ ns (1.13 mm; Fig. 5, left). A residual inter-plane time-walk correlated with the plane charge asymmetry (-105 ns per unit asymmetry; Fig. 5, right) — the brighter plane crosses threshold slightly earlier — is correctable, taking the resolution to a floor of 29.0 ns (1.00 mm). The leading edge is by far the sharpest event estimator: the median strip time gives 70.3 ns per plane, and the trailing (mesh-side) edge is worse still, as expected from longitudinal diffusion.

4.3 C. Absolute event-time resolution and its budget

The absolute leading-edge event time has $\sigma_{68} = 37.7$ ns (Fig. 2, right), an upper limit because it still folds the drift geometry. Decomposing $\sigma_{\text{abs}}^2 = \sigma_{\text{det}}^2 + \sigma_{\text{scint}}^2 + \sigma_{\text{ftst}}^2 + \sigma_{\text{geom}}^2$ with the detector term from (B), the scintillator pair ~ 5 ns, and the fine-timestamp quantization $10 \text{ ns}/\sqrt{12} = 2.9$ ns, leaves a geometry term of ~ 17 ns. **The absolute time budget is dominated by the detector,** not by the trigger: the scintillator and the `ftst` granularity are subdominant.

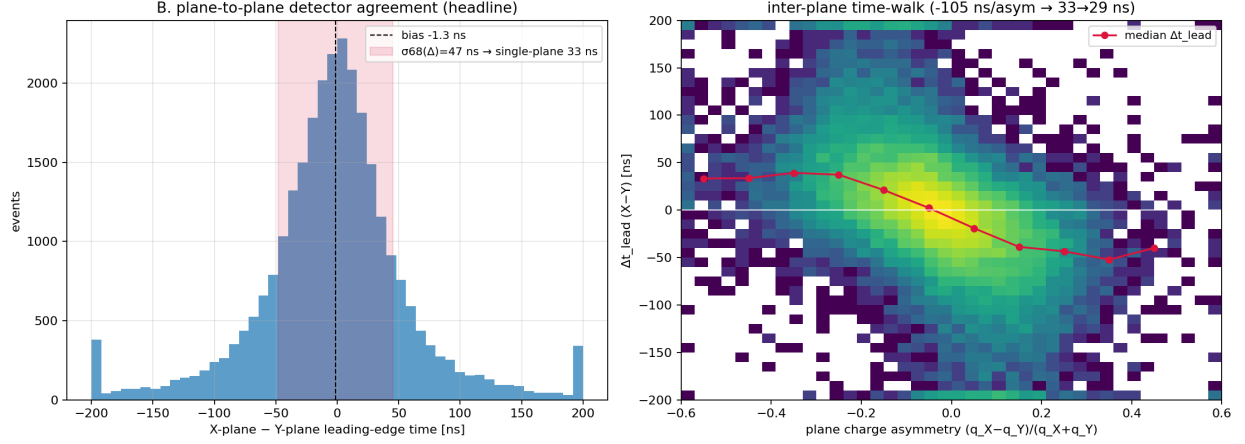


Figure 5: *Left:* X–Y leading-edge time difference; bias ≈ 0 and $\sigma_{68}(\Delta) = 47$ ns $\Rightarrow$ single-plane 33 ns. *Right:* the residual inter-plane time-walk vs plane charge asymmetry; its removal takes the resolution from 33 to 29 ns.

4.4 Time resolution vs drift depth

Binning the plane-to-plane resolution by the event’s leading-edge drift time (Fig. 6) shows σ_t rising from ~ 28 ns (~ 1 mm) for tracks reaching close to the readout to > 150 ns for tracks whose earliest charge is already deep in the gap. The growth is steeper than pure diffusion, reflecting the additional signal-to-noise loss from electron attachment (the same $\lambda \approx 15$ mm attachment that sets the drift-gap story).

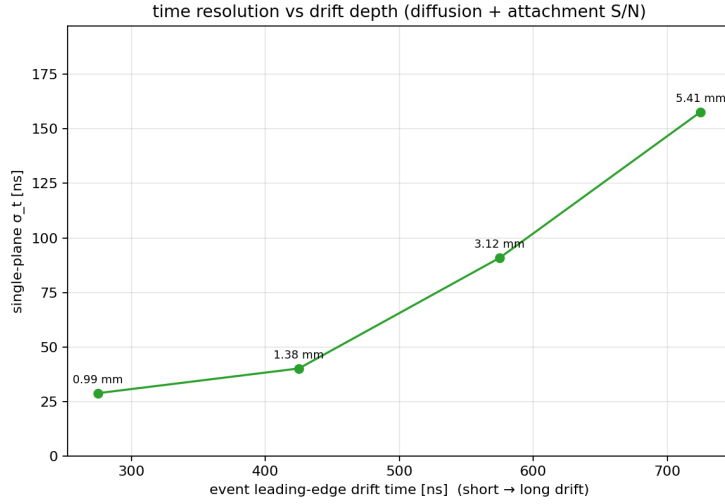


Figure 6: Single-plane detector σ_t vs event drift depth. Short drift (~ 1 mm) is best; the rise with depth is diffusion plus attachment-driven S/N loss.

5 Cross-check: det2 replication

Repeating the analysis on an independent detector (o22_long_det2, 525 V, different FEU pair, 23 894 dual-plane events) reproduces every number within $\sim 10\%$ (Table 1), establishing these as a

design/gas property rather than a per-chamber accident.

Table 1: Time-resolution summary. Drift-distance equivalents use $v_{\text{drift}} = 34.30 \mu\text{m/ns}$.

Quantity	det3 (<code>sat_det3</code>)	det2 (<code>o22_long_det2</code>)	equiv. drift
Single-strip σ_t	38.9 ns	40.5 ns	1.33/1.39 mm
Detector σ_t (leading edge)	33.1 ns	36.8 ns	1.13/1.26 mm
Detector σ_t (walk-corrected)	29.0 ns	34.0 ns	1.00/1.17 mm
Inter-plane bias	-1.3 ns	-2.2 ns	≈ 0
Inter-plane walk	-105 ns/asym	-96 ns/asym	—
Absolute $\sigma_{68}(t_0)$ (UL)	37.7 ns	37.6 ns	1.29/1.29 mm
<code>ftst</code> slope (corr / uncorr)	-0.2/ - 10.2	+0.4/ - 9.6	ns/step

6 Discussion

Time-walk and its interplay with unsharing. Two distinct distortions can bias micro-TPC timing. The dominant one is the resistive-strip RC *charge/time spreading*: charge diffuses to neighbouring strips as delayed copies, and this is removed by the waveform unsharing (deconvolution) already in the analysis chain. The second is per-strip amplitude *time-walk*, a calibration of time vs pulse amplitude. We find the latter is already small (+3 ns/rel-amp, Sec. 4.1) because the reconstruction times on a 30% constant fraction. The correct ordering, should a walk calibration ever be wanted, is: unshare first (it changes the per-strip amplitudes), then time each cleaned strip, then apply a residual walk correction on the *unshared* amplitude. In practice, unsharing is the dominant lever and the walk correction is polish; the residual inter-plane walk we correct in (B) is a small S/N effect, not a large systematic.

What is and is not measured. The absolute event-time resolution is a genuine upper limit (37.7 ns); a fully model-independent absolute number would require pinning the drift geometry per event (e.g. with the telescope), which we avoid to keep the measurement self-contained. The DAQ records no scintillator waveform, so the ~ 5 ns paddle term is taken from the hardware and cannot be measured from these data — but it is provably subdominant. The detector-only plane-to-plane number is the clean, reproducible figure of merit.

Connection to spatial resolution. At ≈ 33 ns the detector times an event to ≈ 1 mm of drift, comparable to the transverse spatial resolution (0.6–0.8 mm). The longitudinal (drift) and transverse precisions are matched: the micro-TPC is not timing-limited relative to its own geometry.

7 Conclusion

The MX17 resistive Micromegas has an absolute time reference on the cosmic bench (scintillator trigger + applied `ftst` phase correction, verified here), and a detector time resolution of ≈ 33 ns per plane (≈ 29 ns after time-walk correction), equivalent to ≈ 1 mm of drift and reproduced on two detectors. The absolute event-time budget is detector-dominated. This closes paper topic 10: the detector’s intrinsic timing is measured; the absolute time-of-flight resolution is quoted as a detector-limited upper limit, with the honest caveat that the scintillator term is external to these data.

Reproduce: `../.venv/bin/python 42_time_resolution.py sat_det3 --veto=50` (and `o22_long_det2`). Outputs under `alignment_tpc_veto50/time_resolution/`. Numbers in `det3_summary.json` / `det2_summary.json`. Reconstruction: `mm_strip_reconstruction/waveform_analysis/src/WaveformAnalyzer.cpp` (timing lines 848-867, `ftst` lines 390-392).